

| Question        |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |     |     |       |       |      |
|-----------------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|                 |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4.              |     |  |  | Any two of following for (1) each <ul style="list-style-type: none"> <li>Use a colorimeter to measure colour changes <u>with time</u></li> <li>Use a pH meter to measure pH changes <u>with time</u></li> <li>Use a gas syringe / collect gas over water to measure changes in volume <u>with time</u></li> <li>Use a balance to measure mass changes <u>with time</u></li> </ul> Award (1) for one correct method but no mention of time |                 | 2   |     | 2     |       | 2    |
| 5.              | (a) |  |  | Presence of a (carbon to carbon) double bond                                                                                                                                                                                                                                                                                                                                                                                              | 1               |     |     | 1     |       |      |
|                 | (b) |  |  | No – because two of the groups on one of the double bonded carbon atoms are the same                                                                                                                                                                                                                                                                                                                                                      |                 | 1   |     | 1     |       |      |
| 6.              |     |  |  | $\text{CuO} + 2\text{HCOOH} \rightarrow (\text{HCOO})_2\text{Cu} + \text{H}_2\text{O}$<br>(ignore state symbols)                                                                                                                                                                                                                                                                                                                          |                 | 1   |     | 1     |       |      |
| Section A total |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1               | 9   | 0   | 10    | 0     | 2    |